

ArtfulSign: A Closed-Loop, Semantics-Grounded Mobile System for Learning Chinese Sign Language

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Abstract

Learning Chinese Sign Language (CSL) is fundamentally an embodied skill learning process that requires repeated physical practice and timely feedback, yet most existing tools remain largely open-loop and focus on passive content consumption. We explore how a closed-loop, semantics-grounded interaction design can better support independent CSL learning, and present ArtfulSign, a mobile system that integrates real-time, on-device recognition feedback with meaning-oriented explanations and lightweight situated scenarios. ArtfulSign organizes learning into short practice loops in which learners perform a sign, receive immediate feedback, and iteratively refine their performance. We will demonstrate ArtfulSign as an interactive CHI demo, where attendees can experience closed-loop embodied practice and learn a CSL sign within minutes.

CCS Concepts

• **Human-centered computing** → **Interaction techniques**; *Accessibility*; Interactive systems and tools; • **Applied computing** → *Education*.

Keywords

Chinese Sign Language, Embodied Skill Learning, Closed-loop Interaction, Practice-based Learning, Mobile Interaction, Accessibility, Semantics Grounding

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1 Introduction

Learning Chinese Sign Language (CSL) is fundamentally an embodied skill learning process. Unlike symbolic or text-based knowledge, CSL is acquired through repeated physical practice involving hand shapes, movements, and spatial configurations. For beginners, simply watching demonstration videos is rarely sufficient: learners often struggle to know whether they are performing a sign correctly, how to refine their movements, and why a particular sign takes its specific form.

However, many existing sign language learning tools emphasize content presentation—such as sign videos, illustrated dictionaries, or browsing-based tutorials—with limited support for observing learners’ performance and providing actionable feedback [1, 2, 6, 10]. As a result, learners are left in a “blind imitation” mode, where mistakes cannot be detected or corrected and progress relies on external instructors or guesswork.

In this work, we explore a different interaction stance: treating CSL learning as a closed-loop, practice-centered, and semantics-grounded process. We argue that effective learning systems should not only present content, but also observe learners’ actions, evaluate performance, and provide immediate, interpretable feedback to structure learning as an iterative practice loop [4, 8]. Moreover, practice should be grounded in semantic understanding—helping learners grasp not only how to perform a sign, but also why it takes its particular form.

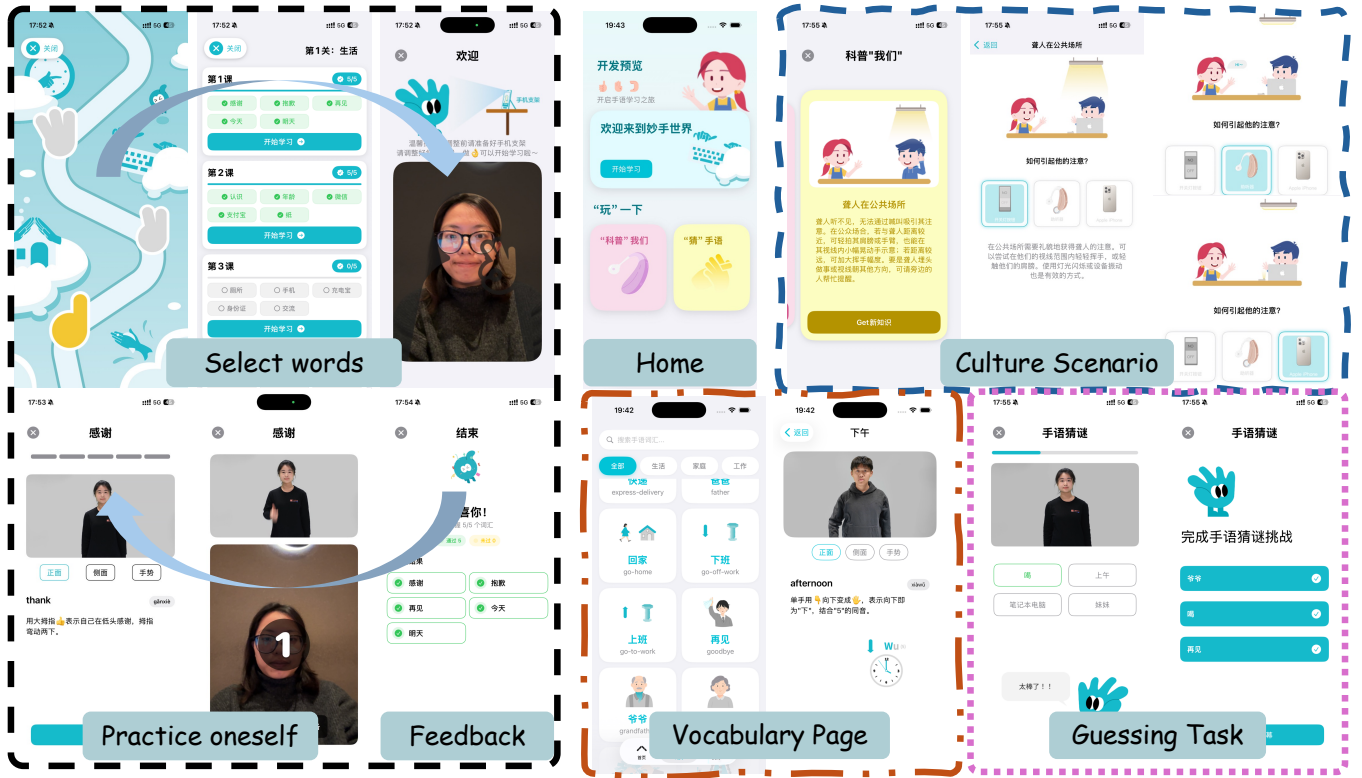


Figure 1: System overview and key interaction screens of ArtfulSign, including the home screen, a vocabulary item page with meaning mapping, the practice feedback interface, a lightweight culture scenario, and a guessing task.

We instantiate this design stance in ArtfulSign, a mobile learning system that supports closed-loop CSL practice with real-time, on-device feedback and meaning-oriented explanations, complemented by lightweight situated scenarios. ArtfulSign has been deployed as a real-world mobile application and supports short, repeatable practice sessions without requiring accounts or network connectivity. In this paper, we present ArtfulSign as an interactive CHI demo that allows attendees to experience closed-loop embodied practice firsthand.

Contributions. We make a system and interaction design contribution by:

- (1) articulating and instantiating a closed-loop, practice-centered interaction design for CSL learning;
- (2) showing how immediate, interpretable feedback can structure short, iterative embodied practice sessions; and
- (3) integrating semantic grounding and lightweight situated context as scaffolding for meaningful, rather than purely imitative, sign language learning.

2 Related Work

Our work connects three strands of prior research: sign language learning tools, embodied and skill learning with feedback, and interactive learning systems that structure practice as iterative loops [4, 8].

Sign language learning tools. Existing CSL and sign language learning applications are predominantly designed as content repositories, such as video dictionaries, phrase collections, or browsing-based tutorials. While some systems introduce lightweight interaction elements (e.g., quizzes, levels, or progress tracking), the core learning experience remains centered on watching and following demonstrations. Learners’ actual performance is rarely observed or assessed by the system, leaving practice fundamentally open-loop. **Embodied and skill learning with feedback.** A substantial body of HCI research shows that learning embodied skills—such as gestures, sports, musical performance, or physical rehabilitation—benefits from timely and interpretable feedback [8]. Prior work explores visual, auditory, and haptic feedback to support self-correction and motor learning [4, 8]. In sign-related contexts, recent systems for sign recognition and translation further demonstrate the feasibility of real-time, camera-based analysis on consumer devices [9].

Interactive learning systems as practice loops. Most sign language learning tools remain open-loop, while many feedback-based embodied learning systems are not designed as everyday, learner-facing practice tools. ArtfulSign bridges this gap by treating CSL learning as a closed-loop, practice-centered interaction process on mobile devices. It integrates immediate performance feedback with semantic grounding and lightweight situated context, turning

learning into short, repeatable practice loops rather than one-way content consumption.

3 Design Rationale: Closed-Loop, Semantics-Grounded Practice

Our design is guided by the view that learning Chinese Sign Language (CSL) is primarily an embodied skill learning process rather than the acquisition of symbolic knowledge [3, 5]. This perspective shifts our focus from content presentation to how interaction structures can support repeated practice, self-correction, and meaningful understanding.

3.1 CSL as Embodied Skill Learning

CSL is produced and perceived through coordinated hand shapes, movements, and spatial configurations. For beginners, progress depends on repeated physical practice rather than passive observation. While videos and illustrations can demonstrate what a sign looks like, they do not help learners judge or refine their own performance. Treating CSL learning as embodied skill learning therefore calls for interactive systems that can support practice and feedback.

3.2 From Open-loop Exposure to Closed-loop Practice

Many existing learning tools implicitly adopt an open-loop structure: the system presents examples, but does not observe or respond to learners' actual performance, leaving error detection and correction to the learners themselves.

In contrast, a closed-loop interaction continuously connects action, assessment, and feedback. The system observes performance, evaluates it, and provides immediate feedback that shapes the next attempt [4, 8]. This perform–assess–adjust cycle turns learning into an iterative refinement process rather than one-shot imitation, which is especially important for CSL, where subtle differences in hand shape, movement, or orientation can determine correctness.

3.3 Beyond Mechanical Repetition: The Role of Semantic Grounding

While closed-loop feedback supports motor refinement, repetition alone does not guarantee meaningful learning. Learners may reproduce a gesture without understanding what it represents or why it takes a particular form. Because many signs are motivated by iconic or metaphorical relationships to their meanings, we argue that practice should be semantically grounded [7]: feedback and repetition should be accompanied by explanations that connect form to meaning, supporting both motor execution and conceptual understanding.

3.4 Design Principles

Based on the above rationale, we derive four design principles that guide ArtfulSign:

P1: Closed-loop practice structure. Learning interactions should be organized as short, repeatable loops of performing, receiving feedback, and refining actions.

P2: Immediate and interpretable feedback. Feedback should be provided in real time and in a form that learners can easily understand and act upon.

P3: Meaning-oriented scaffolding. Practice should be accompanied by semantic explanations that link the form of a sign to its meaning.

P4: Lightweight situated context. Learning should occasionally be situated in simple scenarios that illustrate everyday use without overwhelming the core practice loop.

4 System & Interaction Design: ArtfulSign

ArtfulSign is a mobile learning system that instantiates the closed-loop, semantics-grounded practice principles described in Section 3. Rather than organizing learning around content browsing, the system is structured around short, repeatable practice interactions. Figure 1 shows the main interface and interaction flow.

4.1 System Overview

ArtfulSign is implemented as an iOS application that runs fully on-device and supports offline use. The system centers on a practice-oriented Learning mode, complemented by auxiliary modes for exploration and reinforcement (Recognition, Culture, Guessing, and Dictionary). In typical use, learners enter the Learning mode, select a word, and engage in a short closed-loop practice session lasting about one to two minutes. An optional lightweight personalization step is provided on first launch, without requiring any account or network connection.

4.2 Closed-loop Practice Interaction

The core interaction of ArtfulSign instantiates the closed-loop practice structure (P1) through a four-stage loop: prepare, perform, receive feedback, and retry or pass (Figure 2).

Prepare. The system checks whether the learner is properly framed in the camera view and provides simple on-screen guidance if needed.

Perform. The learner performs the target sign in front of the camera while the system observes the movement in real time.

Feedback. The system provides immediate, interpretable feedback (P2) in the form of a progress indicator and a clear correct/incorrect outcome, prompting a retry when necessary.

Retry or pass. The learner either tries again to refine the performance or proceeds once the system confirms correctness.

This loop turns each vocabulary item into a short, self-contained practice episode and makes iterative refinement the default mode of interaction rather than one-off imitation.

4.3 Semantic Grounding through Meaning Mapping

To prevent blind imitation and support meaning-oriented learning (P3), ArtfulSign integrates semantic grounding directly into the practice flow. For each vocabulary item, the system presents a short explanation and an image or animated illustration that links the form of the sign to its underlying meaning or iconic motivation.

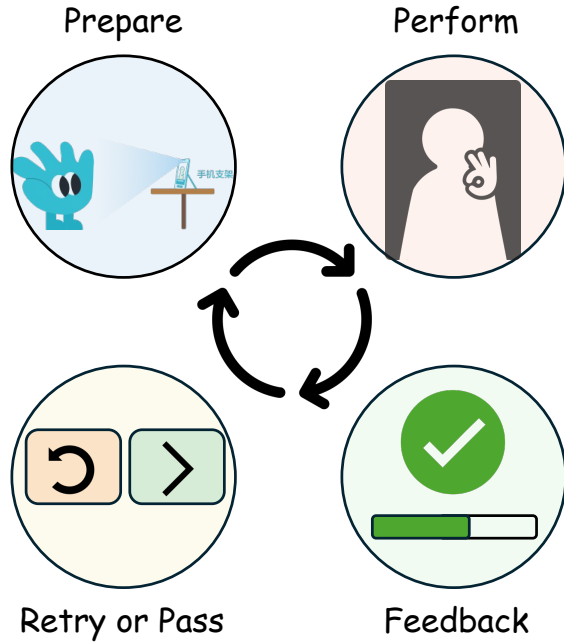


Figure 2: The closed-loop practice interaction in Artful-Sign. Each vocabulary item is practiced through a four-stage cycle—prepare, perform, feedback, and retry or pass—emphasizing immediate, interpretable feedback and iterative refinement.

For example, the sign for “baby” is accompanied by an illustration highlighting the iconic action of holding a baby (Figure 3a). By making such form–meaning relationships explicit, the system helps learners understand why a sign takes a particular shape instead of treating it as an arbitrary gesture to be memorized.

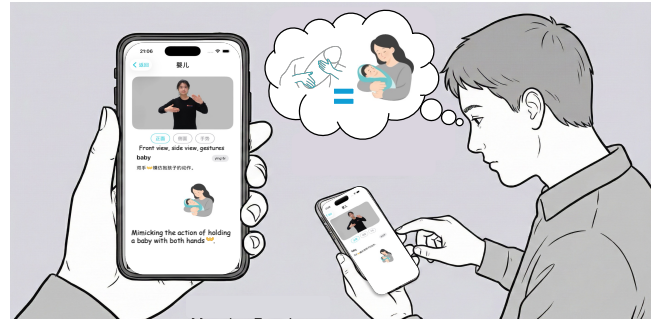
4.4 Lightweight Situated Micro-contexts

In addition to word-level practice, ArtfulSign includes a small set of short, situated scenarios that illustrate how deaf people commonly experience and navigate everyday situations (Figure 3b). These micro-contexts are presented through brief textual explanations and animated illustrations, with optional audio simulations to support hearing learners’ understanding.

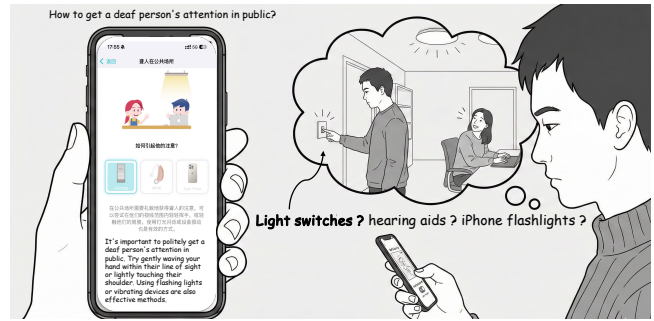
Rather than aiming for comprehensive situational training, these scenarios are intentionally designed as lightweight supplements. They provide pragmatic and cultural context—such as communication challenges and appropriate interaction strategies—while preserving the app’s core closed-loop practice structure focused on individual signs.

5 Discussion and Limitations

ArtfulSign is presented as an interaction design exploration rather than a controlled study of learning outcomes. While the system has been deployed as a real-world mobile application, we do not claim validated learning effectiveness through formal user studies; instead, this demo invites the CHI community to experience and



(a) Semantic grounding through meaning mapping.



(b) Lightweight situated micro-context.

Figure 3: Semantic and cultural grounding in ArtfulSign. (a) Meaning mapping links the form of a sign to its underlying meaning or iconic motivation. (b) A lightweight situated micro-context conveys pragmatic and cultural context through brief text, animated illustrations, and optional audio simulation.

reflect on closed-loop, semantics-grounded practice interactions firsthand.

We note three limitations. First, ArtfulSign prioritizes learnability and semantic understanding over coverage: the current vocabulary set supports short, focused demo interactions but does not aim for comprehensive language instruction. Second, the recognition-based practice loop is sensitive to lighting, occlusion, and camera framing, and performs best with a stable, unobstructed view—which motivates a single-attendee interaction zone in the demo setup. Third, the situated micro-contexts are intentionally brief and selective; they complement rather than replace the core practice loop and could be expanded into longer or more personalized scenarios in future work.

Regarding accessibility and privacy, ArtfulSign runs offline without accounts, and camera input is processed on-device for real-time recognition and is not transmitted. As a primarily visual learning experience, the system does not currently support non-visual access (e.g., for blind users), which remains an important direction for future work.

6 Conclusion

This demo presents ArtfulSign as a concrete exploration of how closed-loop, semantics-grounded interaction design can support

embodied CSL learning in everyday settings. Rather than treating learning as passive content consumption, ArtfulSign structures practice as an iterative cycle of performing, receiving feedback, and refining actions, while grounding these actions in meaning and lightweight situated context. We invite CHI attendees to experience ArtfulSign hands-on, learn a CSL sign within minutes, and reflect on how interactive mobile systems can be designed to better support embodied and meaningful skill learning.

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